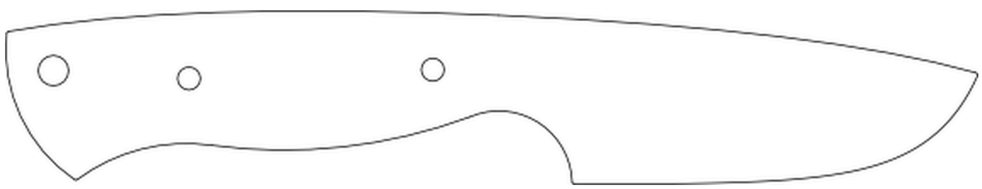
AZ CUSTOM KNIVES

INTRODUCTION TO KNIFE MAKING

AFTER HOURS PROGRAM



THE BADGER

AZ CUSTOM KNIVES

INTRODUCTION TO KNIFE MAKING

AFTER HOURS PROGRAM

AZ Custom Knives

A Slice of Excellence

This program's project knife: the Badger

Disclaimer:
I am teaching one way to make a knife, maybe my way but not the only way!

PROGRAM OVERVIEW

DAY 1:	Profiling
DAY 2:	Rough Grinding
DAY 3:	Heat Treatment
DAY 4:	Handle Work – Part 1 (Preparation & Assembly)
DAY 5:	Handle Work – Part 2 (Shaping & Finishing)
DAY 6:	Sharpening

DAY 1

PROFILING

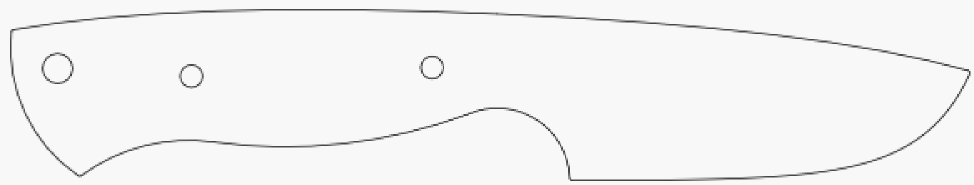
TODAY'S STEPS

1. Choose a shape
2. Transfer design on to desired steel
3. Cut out design
4. Grind to exact shape
5. Mark and drill holes
6. Ensure no sharp edges are present

PROFILING

1 CHOOSE A SHAPE

In this program, we are making the "Badger", a robust fixed-blade knife with a full-tang ergonomic finger-grooved handle and a versatile drop-point blade.



BADGER

NOTES:

2 TRANSFER DESIGN ON TO DESIRED STEEL

We are using 1084 carbon steel, a simple and forging steel, perfect for beginners. Using a Sharpie or scribe, we transfer the blank shape onto the bar stock.

> **Note:** Using a sharpie allows you to be more flexible with your shape. The scribe allows you to be more accurate but requires you to fully grind away the scribe line around the ricasso area.

> **Tip:** Also mark the holes, it will help finding the best placement for drilling later.

NOTES:

3 CUT OUT DESIGN

Using a zip-cut, we are cutting out the rough shape of the knife. Staying away from the line while cutting ensures that we can stay true to the shape of the knife. We will refine the shape with the belt grinder (in the next step) to the exact shape we want.

> **Note:** During this and any following steps we want to make sure we don't scratch up the surface of the blank too excessively. Deep scratches might show in the end product especially in the recasso area.

> **Tip:** To minimize deep scratches, make sure to cut in from the side and not from the top surface, skipping over the surface with the zip-cut is most likely gonna leave marks that we simply not want to deal with.

NOTES:

4 GRIND TO EXACT SHAPE

Using the belt grinder, we carefully remove all excess material that we don't want. This is the point where we learn the belt. How it cuts, where it cuts what to do and what not to do.

In general we want to use the full width of the belt. For more rapid stock removal, using only the edge of the belt can be advantageous but generally rounds off the belt. We are using the flat platen for any of the positive curves and resort to the small wheel and round contact wheel to refine the negative curves.

> **Note:** A used 36 or 60 grit belt is perfect to profile most blanks. A new belt will be more rapid and cooler running than a used one but any mistakes will have greater consequences.

> **Tip:** To minimize scratches, make sure to de-burr the blank on the ~100 grit sand paper on a granite. Do this any time you create and excessive burr. This way you are preventing big metal pieces to get imbedded into the table.

NOTES:

5 MARK AND DRILL HOLES

As we are working with an existing blank, we have the pleasure of simply transferring any of the existing hole onto our blank.

To ensure our holes are exact, we are following these few steps "to a T":

1. Clamp the blank to the template at 2 points.
2. Start the drill and feed the drill bit into the first hole until you hit the steel in you blank.
3. Clamp down the blanks while the drill bit is spinning freely in the hole.
4. Ensure that the drill bit does not "kick" to the side if you move in and out of the hole.
5. Drill the first hole.
6. De-burr the hole on the bottom side.
7. Before removing clamps, put a snug pin into the hole you just drilled.
8. Move the one clamp tip the other side so you can drill the other hole.
9. Repeat the drilling process for any consecutive hole.
10. Unclamp blank from template and de-burr all holes.

> **Note:** Make sure always have have at least 2 point of retention active, this can either be 2 clamps or a clamp and a pin.

> **Tip:** Temporarily clamping the 2 blanks together with a spring clamp allow for a quick set on your screw clamp.

NOTES:

6 ENSURE NO SHARP EDGES ARE PRESENT

For any of the future steps, it is advisable to de-burr all sharp corners, this counts for the circumference as well as all the holes.

> **Note:** Sharp corners can create sources of stress during heat treatment, with de-burring our blank at this point we assure that we are re best prepared for any of the next steps.

> **Tip:** My favourite way to de-burr is to let the blanks be stone washed for a night. The result is a very uniform de-burr all the way around.

NOTES:

DAY 2

ROUGH GRINDING

TODAY'S STEPS

1. Marking the centre of our cutting edge
2. Prep edge for grind
3. Find and grind desired angle onto the blade
4. Grind the bevel
5. Refine grind pattern
6. Ensure symmetry within plunge and grind
7. Ensure no sharp edges are present

ROUGH GRINDING

1 MARKING THE CENTRE OF OUR CUTTING EDGE

With a scribe, drill bit or specialty tool, we are marking the centre of the (soon to be) cutting edge.

> **Note:** Marking 2 parallel lines to mark your cutting edge, makes for a more accurate grind.

> **Tip:** For the best contrast of your scribe, refine scratch pattern on the cutting edge to a minimum of 120 grit and/or apply a marking dye.

NOTES:

2 PREP EDGE FOR GRIND

Using an old belt or disc, we will grind a small bevel onto the cutting edge making sure we don't grind our centre lines away.

> **Note:** This will help save our new belt from losing a fair bit of life in the first few strokes. We are basically opening the approach angle of the grit hitting our steel edge, minimizing the amount of sharp grit being sheared away right off the bat.

> **Tip:** Grind halfway to the marked line. Getting too close to the marked line leaves little material to finalize our angle.

NOTES:

3 FIND AND GRIND DESIRED ANGLE ONTO THE BLADE

Choose your angle and position of plunge cut. With a 36 grit belt, carefully remove excess material from the blade. Readjusting our angle might be necessary if we approach one point /line of our bevel more rapidly (percentually) than the other.

> **Note:** Using any type of jig will help restrict unwanted movement and therefore unwanted stock removal. A plunge guide jig will do this with one direction. A bevel jig will do this for 2 out of the 3 directions.

> **Note:** When using a bevel jig, make sure your table is completely clean and smooth. Any shavings that have landed on the table will need to be removed, otherwise it can create deep scratches into the table and the bottom of the jig. Periodically clean your table with Scotch-brite and the bottom of the jig with sandpaper on a granite.

> **Tip:** For extra smooth travel with the jig, apply silicone spray to the table and bottom of your jig.

NOTES:

4 GRIND THE BEVEL

Stroke by stroke against the belt, we are creating our bevel until we are right on the line we drew for ourselves.

> **Note:** Knowing which style and grit of belt suits your way of grinding the most is extraordinarily helpful to be efficient in time and with belt utilization.

> **Tip:** Starting out with a brand new belt once the correct bevel has been found can be very satisfying and rapidly lets you approach the next step.

NOTES:

5 REFINE GRIND PATTERN

Using a 60 grit belt, refine the scratch pattern. Ensure both sides are equal and the plunge is symmetrical. Move up in grit to a 120 grit belt if a smoother finish is wanted and reasonable to achieve.

> **Note:** We will want to leave this blade at a rough ground finish with no refinement after the heat treat. In this case, a finer surface finish is reasonable.

> **Tip:** Hand sanding the bevel may be tedious, but it yields great results.

NOTES:

6 ENSURE SYMMETRY WITHIN PLUNGE AND GRIND

For the best potential success within the heat treat, we must ensure that our grinds are equal; the plunge also needs to be symmetrically shaped and positioned.

> **Note:** If we grind too far on the one side, we will need to mimic that on the other side. Symmetry is more important than having one side exactly how you want it.

> **Tip:** Chasing the plunge, bevel and flats equal symmetry is best done during the process of refining the grind pattern.

NOTES:

7 ENSURE NO SHARP EDGES ARE PRESENT

Ensure all new sharp edges created during the grinding process are de-burred.

> **Note:** Realize this is the last step before the heat treat and that all modifications done to the blank from here on out will be done when the steel as at a hardened state.

> **Tip:** My favourite way to de-burr is to let the blanks be stone washed for a night. The result is a very uniform de-burr all the way around.

NOTES:

DAY 3

HEAT TREATMENT

TODAY'S STEPS

1. Ensuring blank is clean and there are no sharp edges present
2. Heat up blank to desired temperature
3. Quenching
4. Tempering

HEAT TREATMENT

1 ENSURING BLANK IS CLEAN AND THERE ARE NO SHARP EDGES PRESENT

Ensure blank is free of any sharp edges and all the previous steps have been completed. Wipe blank clean with acetone before proceeding with the heat treatment.

> **Note:** Sharp edges can create stress lines in the steel that potentially can cause a crack or failure in the steel.

> **Note:** Any oil on the surface will bake onto the steel during heat treat.

> **Tip:** Take your time to properly prepare for the heat treat; this is a crucial point in the making of our knife.

NOTES:

2 HEAT UP BLANK TO DESIRED TEMPERATURE

In this program we are using our heat treat kiln to heat up our blanks. The following steps of heat treat are appropriate for 1084 steel.

Normalization (optional)

If the 1084 steel has an uneven or non-uniform structure, you may want to normalize it first. Heat the steel to around 1650ºF (989ºC) for 10 to 15 minutes and let it cool in still air. Repeat with a second cycle at 1500ºF (815ºC) and a 3rd cycle at 1350ºF (732ºC). Normalizing helps refine the grain structure and relieve internal stresses.

Austenitizing

Heat the 1084 steel to a temperature between 1450-1500ºF (790-815ºC) to convert its microstructure into austenite. Hold at this temperature for a sufficient amount of time to ensure the entire piece reaches the same temperature.

> **Action:** Heat up the kiln to 1400ºF and place blank into kiln. Let kiln reach 1465ºF.

> **Note:** With our thickness of 1/8" we will assume that the entire piece is hot to the core and fully transformed once the kiln reaches temperature.

NOTES:

3 QUENCHING

After austenitizing, the steel needs to be rapidly cooled to transform the austenite into martensite – a hard and brittle microstructure. Quenching involves immersing the heated steel in a quenching medium such as oil, water, or brine. Quenching in oil results in a less abrupt and softer quench, while water or brine will provide a more severe and harder quench.

> **Action:** Open the kiln, grab the blank by its handle with tongs, and move the blank across a magnet to ensure it is non-magnetic before submerging it in Park50 oil. Move the blank forward and backwards as far as the container will allow you to.

> **Important Note:** Do not move the blank side-to-side, nor twist it. Any other motion toward one side will greatly increase the potential for warping.

> **Danger Note:** Plunging with a glowing piece of steel into oil can ignite the oil and create a flare-up of fire as well as hot oil splatter. Be sure to use PPE and position yourself appropriately.

> **Tip:** Always make sure to prepare all your steps beforehand. Place the oil in a convenient location and prepare tongs, rags and any PPE needed.

NOTES:

4 TEMPERING

The martensite produced during quenching is too brittle for most applications, so tempering is necessary to reduce hardness and increase toughness. For 1084 steel, tempering is usually performed between 350-600ºF (175-315ºC). The higher the tempering temperature, the lower the hardness but higher the toughness.

For a balance between hardness and toughness, temper at around 400-450ºF (200-230ºC). For a softer, more shock-resistant steel, temper at around 550-600ºF (290-315ºC).

> **Action:** Ensure to clean the blank of any oil and place into oven or kiln at 400ºF for a duration of 2 hours. Take it out and let it come to room temperature before repeating this process one more time.

> **Note:** At 400ºF twice for 2 hours, our steel should have a hardness of 60-61 HRC, which is comparable to a good quality file.

> **Tip:** If the temper happens in an oven or toaster oven, keep an eye on the actual temperature of the steel as those ovens seem to fluctuate greatly in temperature. A temperature gun can help greatly to accurately pinpoint the temperature of the steel.

> **Pro Tip:** Knowing which temperature corresponds with which temperature in the most common steels can help knowing the range of temperature you are dealing with.

NOTES:

TEMPERING COLOURS

The colour of the steel surface changes as it is heated to different temperatures – use as a visual guide:

Normalised	176°C / 349°F	204°C / 399°F	227°C / 441°F	249°C / 480°F	260°C / 500°F	282°C / 540°F	310°C / 590°F
Bright	Yellow	Pale Straw	Dark Straw	Brown	Purple	Blue	Blue-Gray

NOTES:

DAY 4

HANDLE WORK

PART 1 – PREPARATION & ASSEMBLY

TODAY'S STEPS

1. Choose your material
2. Cut out scales
3. Drill holes through the scales
4. Shape scales
5. Cut pins and tubes
6. Prepare for epoxy
7. Epoxy scales onto handle

HANDLE WORK – PART 1

1 CHOOSE YOUR MATERIAL

In this program, we are using G10 – a composite material made from layers of fibreglass fabric impregnated with epoxy resin and compressed under high pressure to form a dense, strong, and rigid sheet.

Advantages: High strength-to-weight ratio, electrical insulation, chemical resistance, dimensional stability, low water absorption, and good machinability.

> **Danger Note:** G10 becomes dangerous when glass fibres are broken out of its bond either with grinding/sanding or machining. Use a respirator with dust filter to protect your respiratory system.

> **Danger Note:** G10 can cause irritation of the skin. Be sure to clean yourself and your surroundings whenever you step away from the station.

NOTES:

2 CUT OUT SCALES

Mark and cut 2 appropriate pieces of G10 to size with the tile saw or band saw.

> **Note:** G10 is extremely abrasive. Although easily machinable, it is very hard on any cutting tool or abrasive you use to shape G10.

> **Tip:** Wear long sleeves and keep a vacuum close. Bonding the dust in water limits the risk of G10 becoming airborne.

NOTES:

3 DRILL HOLES THROUGH THE SCALES

To ensure our holes are exact, we are following these few steps "to a T":

1. Clamp the scale to the blank at 2 points.
2. Start the drill and feed it into the first hole until you hit the G10.
3. Clamp down the blank and scale while the drill bit is spinning freely in the hole.
4. Ensure the drill bit does not "kick" to the side.
5. Drill the first hole.
6. De-burr the hole on the bottom side.
7. Before removing clamps, put a snug pin into the hole you just drilled.
8. Move the clamp to the other side so you can drill the other hole.
9. Repeat for any consecutive hole.
10. Unclamp scales from blank and de-burr all holes.
11. Repeat these steps drilling the second scale.

> **Note:** Make sure to always have at least 2 points of retention active – this can either be 2 clamps or a clamp and a snug pin.

NOTES:

4 SHAPE SCALES

Pin one scale onto the handle. Using a scribe or Sharpie, mark the shape of the handle onto the G10. Figure out what shape you want the front of the handle to be; mark it and transfer that shape onto one pinned scale.

Cut and/or grind one scale about 1/8" oversized from the handle but grind accurately to the mark at the front of the handle. Pin the 2 scales together with a snug pin and cut/grind the second scale to the same shape.

Refine the front of the handle to a fine finish; router the desired profile onto it. Hand-sand the front to final finish.

> **Note:** Finishing the front is an important step as we won't be able to safely work on the front once we have the scales epoxied onto the handle.

> **Danger Note:** Routering scales can be very dangerous and needs to be executed with a very high level of awareness and all safety measures in place.

> **Tip:** Keep using the sandpaper on the granite to maintain a flat surface during cutting and grinding.

NOTES:

5 CUT PINS AND TUBES

Measure the handle thickness by measuring the 2 scales including the blank. Using a hacksaw or zip-cut, cut all pins and tubes of the preferred material to 1/8" – 3/16" longer than the thickness of the handle. De-burr all pins and tubes on both sides, ensure no excessive chamfer.

> **Note:** If G10 pins are cut, make sure to turn the pin while cutting as the G10 likes to break out if cut from one side.

> **Tip:** Cutting multiple pins is done pretty quickly and can save some time on your next project.

NOTES:

6 PREPARE FOR EPOXY

1. The blank

Take your blank and mark where the front of handle goes. Using 36 grit belt on a 2" wheel, roughen up the inside of the handle on both sides. Stay away from the line and outside of the handle as much as you can while making sure there are deep scratches as far out as possible.

> **Note:** Be careful around the front of the handle, any scratches outside of the marked line will be visible in the finished product.

2. Scales

Using a rough 80 grit sandpaper, roughen up the G10 systematically until there are no shiny spots detectable anymore.

> **Note:** Do not round the edges, especially at the front or the scales, any severe rounding of the G10 will be visible.

3. Pins and tubes

Scratch up the surface of the pins and tubes with a rough sandpaper. Grind in 3-4 small notches in the centre of all pins and tubes; this will help retain them when the epoxy sets.

> **Tip:** Using a cordless drill and spinning the pins and tubes while carefully placing 80 grit sand paper onto the pin can yield a quick and proper result.

4. All elements

Ensure that all pieces fit perfectly together and the pins slide through the holes with ease. Clean all elements with acetone and ensure they are free of any oil, dust or fibres.

NOTES:

7 EPOXY SCALES ONTO HANDLE

Before epoxy comes preparation! Make sure all your pieces are properly de-burred, cleaned and free of oils and dust.

Lay out all your pieces on a piece of paper. Prepare your gloves, clamps, spread stick, acetone, Q-tips and some rags/towels.

Measure out the correct ratio of epoxy and hardener and mix thoroughly for 1 minute.

Apply a thin layer of epoxy covering 100% of all surfaces subject to be epoxied. Ensure epoxy find its way around the walls of the holes and around the pins.

Start with one scale and insert the pins, then layer the blank on top by sliding it down through the pins in the handle followed by the last scale.

Ensure that we have epoxy squeezing out all the way around as we start to clamp the scales with pressure distributed evenly throughout the handle. Ensure that the pins are sticking out of the scales on both sides evenly.

Set up the knife with the tip up. With acetone soaked Q-tips, start to clean the epoxy off the front of the handle where it meets the ricasso. Ensure that no epoxy is present on the blade, ricasso and front of the handle including the radius.

Let the knife sit with the tip up until the epoxy is fully cured.

> **Note:** We can use slow or fast setting epoxy depending on time restrictions. This will dictate how quickly we need to execute each of these steps.

> **Tip:** Letting a tiny bit of epoxy squeeze out in front of the handle, this will form a tiny bead of epoxy, making sure we have no voids where debris can settle.

NOTES:

DAY 5

HANDLE WORK

PART 2 – SHAPING & FINISHING

TODAY'S STEPS

8. Grinding pins down
9. Ensure flatness of scales
10. Grind scales down to the tang
11. Router profile on scales
12. Hand sand scales to finish
13. Ensure no sharp corners are present

HANDLE WORK – PART 2

8 GRINDING PINS DOWN

Remove potentially big pieces of epoxy from the flats of the scales without excessively scratching the scales. Using the sandpaper granite, sand down the pins until they are flush with the scales.

> **Note:** Leaving too much epoxy on the scales, especially when the epoxy is not fully cured, will gunk up the sandpaper and is less efficient than cutting off excess epoxy.

> **Note:** Ensure that you are parallel to the sandpaper when you do this. Grinding down the pins and scales at an angle might leave visible marks in the scales and/or can lead to issues later.

> **Tip:** Do not do this step with the grinder, a few minutes of hand work at this time can save a lot of time later if one wishes to have a refined handle.

NOTES:

9 ENSURE FLATNESS OF SCALES

Having our flats parallel, square and flat is imperative for the next step. Using the sandpaper granite, sand the scales flat and parallel.

> **Note:** If step 8 was done correctly, there is no need to do anything.

> **Tip:** Spending the time to do this correctly will enable us to be efficient and precise in our next step.

NOTES:

10 GRIND SCALES DOWN TO THE TANG

Having a completely flat and perpendicular plane to grind our circumference is optimal and will result in efficiency and actuary.

Cover your blade with masking tape to create a buffer and protect it from unwanted scratches. Using the belt grinder, a table and our flat surface on the scales, grind down the scales until you hit the tang/steel.

Start with the positive curves and refine the scratch pattern until desired finish. Continue with the negative curves by using our small wheel and round contact wheel and refine to desired finish. Once satisfied with the finish, using a small wheel and a 600 grit belt, knock off any sharp corners created during the grinding process.

> **Note:** Using a 36 grit to "find" your tang can be quick but can lead to excessive grinding. Using a 60 grit will be much slower and still will have to be done with extreme care.

> **Note:** Grinding excessively on the tang will change our shape and requires extra time to refine.

> **Tip:** When taping the blade, don't try to be nice and flat, make sure you've got some wrinkles to create a functioning buffer.

NOTES:

11 ROUTER PROFILE ON SCALES

Ensure that our flats are free of debris and carefully router desired profile onto the rest of the handle.

> **Danger Note:** Routering scales can be very dangerous and needs to be executed with high level of awareness and all safety measures in place.

> **Note:** A radius will feel nice and smooth while a chamfer will leave a noticeable facet around the handle.

> **Tip:** Taking multiple steps to cut the full depth of the profile can be advantageous.

NOTES:

12 HAND SAND SCALES TO FINISH

Using the sandpaper granite, ensure there are no deep scratches visible. Clamp your blade into a blade vise and start to refine the router marks. Using a finer and finer sandpaper, continue refining to desired finish.

> **Note:** Routering G10 will tear out the glass fibres and leave little tear-outs rather than a smooth finish. Using a 120 sandpaper for the first step will give you a good potential to get all the marks out.

> **Note:** Only touching the routered part ensures efficiency and accuracy.

> **Tip:** G10 sands very nicely with water. Begin introducing water into your sanding at or after 220 grit, while making sure that the water does not dry onto the blade.

> **Tip:** Any tears in the G10 that were not sanded down will show up as white speckles and only do so when the G10 is dry. To see them during wet sanding, dry your piece.

NOTES:

13 ENSURE NO SHARP CORNERS ARE PRESENT

Feel your almost finished knife and make sure you don't feel any sharp corners or edges throughout the scales and steel, especially around where your fingers generally meet the steel. Remove any and all unwanted burrs, sharp corners or edges.

> **Note:** Removing burrs with the belt grinder can be quick but requires good control and diligence.

> **Tip:** Clean your knife with soap and water.

NOTES:

DAY 6

SHARPENING

TODAY'S STEPS

1. Remove material to create a cutting edge at desired angle
2. Create a burr along the cutting edge
3. Refine grind pattern
4. Ensure a burr along the whole cutting edge at desired finish
5. Remove burr
6. Test sharpness

SHARPENING

Sharpening the angles of tools is an essential practice to maintain their cutting efficiency and prolong their lifespan. Different tools require specific angles for optimal performance, and sharpening them correctly is crucial.

Common Sharpening Angles

TOOL	ANGLE
Knives	18–25° per side (Japanese-style: 15–18°)
Chisels	25–30° (paring: ≤20°)
Axes / Hatchets	25–35°
Scissors / Shears	20–30°
Mower Blades	40–45°
Drill Bits	~118° for metal
Plane Blades	25–30°

NOTES:

1 REMOVE MATERIAL TO CREATE A CUTTING EDGE AT DESIRED ANGLE

Depending on thickness of cutting edge, choose an appropriate grit. In our case a 120 grit should suffice. Using the belt grinder, grind away material from the cutting edge at the desire angle. Do this from both sides equally to meet in the middle of the cutting edge.

> **Note:** A 18-20 degree cutting edge is appropriate for our steel used and purpose of knife.

> **Danger Note:** Grinding away from yourself is safer and the preferred way of doing this. Letting the belt run toward you and the blade can be dangerous yet yield in good results.

> **Tip:** In this "rough grind" stage, chose a slightly bigger angle to remove material than the final angle desired. (19-20 degree if you like to end up an 18 degree)

NOTES:

2 CREATE A BURR ALONG THE CUTTING EDGE

Ensure that you are grinding equally to the centre until you see a burr on the entire cutting edge.

> **Note:** Keep the burr small at this stage ensures that we do not alter the shape of the blade.

> **Tip:** This grind will decide the general shape of your cutting edge, so make sure you are happy with this step before you proceed. Reshaping with smoother grits is not advised.

NOTES:

3 REFINE GRIND PATTERN

Using finer belts, refine the cutting edge and burr until desired finish is reached. Ensure full cutting edge has been chased and refined at the desired angle before refining further. Ensure to chase both sides equally to keep the cutting edge in the centre.

> **Note:** When a burr from the previous grind has been removed, it might fall off and a new smaller burr will become visible.

> **Tip:** Keep away from the tip! The tip is gone so quick that it makes sense to retain the original rough grind at the very tip until the last grit.

NOTES:

4 ENSURE A BURR ALONG THE WHOLE CUTTING EDGE AT DESIRED FINISH

Ensure full cutting edge shows a refined burr along the full length of the cutting edge including the tip and heel. Ensure the cutting edge is in the centre.

> **Note:** This is your final grind on the cutting edge, make sure your happy with it.

> **Tip:** Bringing your finish to a ~800 grit finish at a 18 degree angle will leave you with a bit of tare at a razor sharpness.

NOTES:

5 REMOVE BURR

Removing our burr is the last final step to our razor sharp edge. "Lipping" the burr from one side to the other will eventually break it off and leave us with the desired cutting edge.

Using a leather strop, stroke the cutting edge at a 30-40 degree angle along the full cutting edge, lipping the burr towards one side, then the other. Repeat this process until your burr fall off. Ensure the full burr has been removed.

> **Note:** Keep your strop clean! As the burr starts to fall off, it will leave its pieces on the strop, this can cause nicks or scratches in the cutting edge.

> **Tip:** A couple of full strokes starting from either end of the cutting edge should ensure that the burr is lipped all the way before switching to the other side.

> **Tip:** If no leather strop is available for stropping, many things can aid as a substitute: belt, pants, hand (be careful)...

NOTES:

6 TEST SHARPNESS

Ensure the full cutting edge has the desired cutting ability. From tip to heal you should feel the same roughness/smoothness along the cutting edge.

> **Note:** If any sharpness test fails, repeat appropriate steps in the sharpening process.

> **Note:** A few more strokes on the strop might be all that's need if a sharpness test fails.

> **Tip:** A paper test, finger nail test and shave test should be able sufficient to ensure we have the appropriate cutting edge on our blade.

NOTES: